Solving Numerical Problems

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 43 minutes · Total marks: 43

KEY RULES

Read carefully	Underline the numbers, units and what is asked
Units	Convert all quantities to the same unit before calculating
Speed	$\text{speed} = \dfrac{\text{distance}}{\text{time}}$; $\text{density} = \dfrac{\text{mass}}{\text{volume}}$
Percentage change	$\dfrac{\text{change}}{\text{original}} \times 100$
Show working	Write every stage; give units in the answer

WORKED EXAMPLE 1

[3 marks]

A car travels 150 km in 2 hours 30 minutes. Work out its average speed.

Step 1	Change the time to hours	$2\text{ h }30\text{ min} = 2.5\text{ h}$
Step 2	Speed = distance ÷ time	$150 \div 2.5 = 60$

Answer: 60 **km/h**

WORKED EXAMPLE 2

[3 marks]

2.4 kg of flour costs $3.60. Work out the cost of 1.5 kg.

Step 1	Cost of 1 kg	$3.60 \div 2.4 = 1.50$
Step 2	Cost of 1.5 kg	$1.50 \times 1.5 = 2.25$

Answer: $2.25

WORKED EXAMPLE 3

[3 marks]

Change 72 km/h into m/s.

Step 1	Kilometres to metres	$72\text{ km} = 72\,000\text{ m}$
Step 2	Hours to seconds	$1\text{ h} = 3600\text{ s}$
Step 3	Divide	$72\,000 \div 3600 = 20$

Answer: 20 **m/s**

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Solving Numerical Problems

1 A shop sells boxes of almonds.

12 of these boxes cost a total of 708 dirhams.

Work out the cost of 5 of these boxes.

..... dirhams

(Total for question = 2 marks)

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2 Nav makes bracelets using cord.

Nav has a 6 metre length of cord.
Each bracelet needs 17.5 cm of cord.

Work out the greatest number of bracelets that Nav can make.

.....

(Total for question = 3 marks)

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3 Yuan sells fudge in small bags and in large bags.

Small bag	Large bag
Weight 150 g	Weight 400 g
Cost £1.80	Cost £5

Work out which bag is the better value for money.

Show your working clearly.

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4 Write a number in each box so that each calculation is correct.

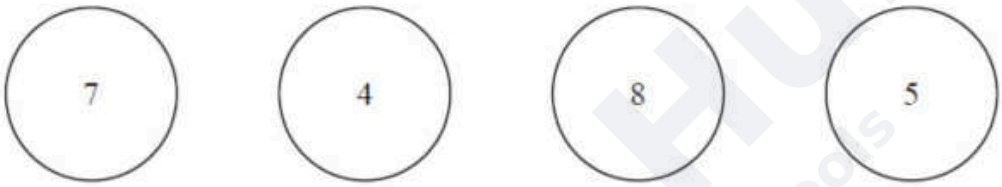
(a) 475 + ☐ = 1200

(1)

(b) 180 × ☐ = 5760

(1)

Here are four discs.
There is a number on each disc.



Using each disc only once,

(c) (i) write down the smallest number that can be made,

.....

(1)

(ii) write down the largest **even** number that can be made.

.....

(1)

(Total for question = 4 marks)

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5 The normal price of a calculator is £26.80

Amy gets $\frac{1}{4}$ off the normal price with her student discount card.

Amy buys 2 of these calculators.
She pays with a £50 note.

How much change should Amy receive?

£.....

(Total for question = 4 marks)

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6 The diagram shows a classroom wall in the shape of a trapezium.

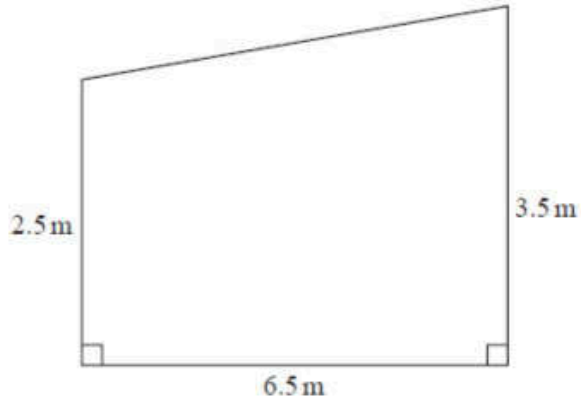


Diagram **NOT** accurately drawn

Dion wants to paint the classroom wall completely twice.
He knows that each tin of paint will cover 12 m^2

He is going to have to buy all the paint he needs.

Work out the least number of tins of paint that Dion will need to buy.
Show your working clearly.

.....

(Total for question = 4 marks)

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CHALLENGE QUESTIONS

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7 The number of runners in the London Marathon on 25th April, 2010 was 37 527.

Work out an estimate for the number of these runners whose birthday was on that day.

.....

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8 **6** Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket **A**	ticket **B**
$\frac{1}{6}$ off normal price	20% off normal price

The sale price of ticket **A** is 140 euros.
The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

(Total for Question 6 is 4 marks)

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9

4 Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

..... %

(Total for Question 4 is 5 marks)

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10 **13** Antoine is going on holiday.
He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$ .....

(Total for Question 13 is 5 marks)

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11 **14** The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

(Total for Question 14 is 5 marks)

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TOTAL FOR PAPER IS 43 MARKS

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